

■ CONCEPT NOTE

AI-Powered Flood Early Warning System

Nyando River Basin · Kisumu County, Kenya

Applicant:	James Onyango Koero — ML Engineer, Kisumu, Kenya
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GitHub:	github.com/jameskoero/nyando-flood-ai
Advisors:	Prof. Johan Loeckx (VUB AI Lab) · Prof. Samuel Liyala (JOOUST)
Date:	May 2026
Funding Request:	USD 25,000 – 150,000 (scalable)

1. Problem Statement

The Nyando River Basin floods almost every April–May rainy season, affecting **50,000–200,000 people annually** across Kisumu, Kericho, and Nandi Counties. Floods destroy crops worth **KES 500M+ per year**. Current early warnings arrive fewer than 6 hours before flooding — insufficient for safe evacuation. No open, locally-trained AI system exists for this basin. *An independent published study (Awino & Machanda, 2024) on this exact watershed confirms both the problem and the feasibility of the ML approach.*

2. Solution — What We Have Built

Component	Status	Evidence
GradientBoosting Model	■ Trained	AUC-ROC 0.9915 · F1 0.9504 · CV 0.9905±0.0011
FastAPI Backend	■ Live	nyando-flood-api.onrender.com/docs
Training Dataset	■ Open	5,000 points · 6 satellite features · CC-BY-4.0
GitHub Repository	■ Public	github.com/jameskoero/nyando-flood-ai
CI Pipeline	■ Green	15 automated tests pass on every commit
React Dashboard	■ Dev	Target Q3 2026 — Leaflet.js choropleth map
SMS Alerts	■ Planned	Africa's Talking API · 10,000 households

3. Model Performance

Metric	Score	Interpretation
AUC-ROC	0.9915	Near-perfect discrimination: flooded vs. not flooded
F1-Score	0.9504	High balance — minimises both false alarms and missed floods
Precision	0.9437	94.4% of HIGH/CRITICAL alerts are genuine flood events
Recall	0.9571	95.7% of real flood zones correctly identified
CV AUC (5-fold)	0.9905 ± 0.0011	Stable across spatial folds — generalises well

4. Data Sources — 100% Open, Non-Personal

Feature	Source	License
Elevation, Slope	NASA NASADEM	Public Domain
3-Day Rainfall	CHIRPS v2 (UCSB)	CC-BY-4.0
Flood Labels	Sentinel-1 SAR (ESA Copernicus)	Free Copernicus
Soil Clay 0-5cm	ISRIC SoilGrids	CC-BY-4.0
River Distance	HydroSHEDS / OpenStreetMap	ODbL
Land Cover	ESA WorldCover 2021	CC-BY-4.0

Privacy & Compliance: All data is satellite-derived. Zero personal or household data. Fully compliant with Kenya Data Protection Act 2019 and GDPR.

5. Impact Metrics

Metric	v1.0 (Current)	v2.0 (With Funding)
Lead time	72 hours	120 hours (WARMA gauges)
Wards covered	42 (Nyando sub-county)	All Kisumu County (7 sub-counties)
Population	50,000+	500,000+
Alert channels	API + Dashboard	+ SMS 10,000 households
Basins	Nyando	+ Tana + Nzoia (Kenya)

6. Budget

Option A — Core Completion (USD 25,000 · 12 months)

Item	USD
Developer time (12 months)	12,000
React dashboard + SMS	4,000
WARMA gauge API access	2,000
Cloud hosting (12 months)	1,500
Field validation workshops ×3	2,500
Publication + conference	1,500
Contingency (10%)	1,500
Total	25,000

7. Team

James Onyango Koero — Lead Developer. B.Sc. Physics & Mathematics, Moi University (2012). Self-taught ML engineer with 5 deployed projects: Nyando Flood AI (AUC 0.9915), AfriSalaries (8-country salary platform), Loan Risk (Gini 0.74, Basel III), Titanic Analysis (AUC 0.9761). Kisumu, Kenya.

Prof. Johan Loeckx — Academic Advisor. VUB AI Lab, Belgium. 40+ years AI research. Peer-review and academic supervision.

Prof. Samuel Liyala — Local Academic Advisor. JOOUST, Kenya. Field access.

8. SDG Alignment

Framework	Alignment
SDG 13 — Climate Action	Direct: AI flood adaptation
SDG 11 — Sustainable Cities	Ward-level resilience planning
Sendai Framework Priority 1	Understanding disaster risk through data
Kenya National Adaptation Plan	Flood priority action 3.2

"The Nyando Basin floods every year. The communities know it is coming. They just do not know when. This project gives them 72 hours to prepare."

— James Koero · Kisumu, Kenya

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